# TIE-COMPLETE NEAREST-SIGNATURE TRANSFER FOR DEPTH-ONE QAOA

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**Abstract.** Repeated graph descriptors create exact distance ties in parameter-transfer libraries. Ordinary nearest-neighbor prediction resolves a boundary tie by row order, making the result dependent on data serialization. We introduce a tie-complete rule that averages every training target at or inside the $k$th-neighbor radius. The rule is permutation invariant and obeys a local Lipschitz error bound. A commutator argument converts its angle error into a depth-one MaxCut energy certificate. The evaluation authenticates Stanford Gset G1 by SHA-256, extracts 48 ten-vertex neighborhoods, and computes noiseless 81-point-grid QAOA references. Thirty-four graph signatures and five angle pairs occur. Under four row-held-out folds, tie completion reduces mean angle error from $1.37 \times 10^{-3}$ for ordinary nearest neighbors to $7.23 \times 10^{-4}$; mean relative regret is $3.09 \times 10^{-4}$ for both. Under the primary signature-grouped protocol, training and test signatures are disjoint. Both neighbor rules then have mean angle error $7.80 \times 10^{-3}$ and regret $2.67 \times 10^{-3}$, compared with $7.99 \times 10^{-2}$ and $1.64 \times 10^{-2}$ for a constant predictor. Thus tie completion removes an algorithmic ambiguity and improves duplicated row splits, but it does not improve extrapolation beyond unseen signatures. The complete negative result is part of the claim.

**Key words.** QAOA, parameter transfer, nearest neighbors, tie invariance, MaxCut

**1. Transfer rule.** For each graph neighborhood, let $x_i \in \mathbb{R}^d$ be a standardized descriptor and let $y_i = (\gamma_i, \beta_i) \in [0, \pi] \times [0, \pi/2]$ be a depth-one reference. Given a query $x$, order squared Euclidean distances $d_i(x) = \|x - x_i\|_2^2$ and let $\rho_k(x)$ be the $k$th smallest distance. The tie-complete neighbor set and prediction are

$$\mathcal{N}_k(x) = \{i : d_i(x) \le \rho_k(x)\},$$
$$\widehat{y}(x) = \frac{1}{|\mathcal{N}_k(x)|} \sum_{i \in \mathcal{N}_k(x)} y_i. \tag{1.1}$$

The value of $k \in \{1, 3, 5, 9, 15\}$ is chosen on a development partition. All points tied at the selected radius are retained. No target label from the test partition participates in scaling, tuning, or prediction.

**Theorem 1** (invariance, local error, and cost)**.** *The prediction in Eq.* (1.1) *is invariant under every permutation of the training rows. Suppose a target map $y(\cdot)$ satisfies*

$$\|y(x_i) - y(x)\|_2 \le L \|x_i - x\|_2 \quad \text{for every } i \in \mathcal{N}_k(x). \tag{1.2}$$

*Then*

$$\|\widehat{y}(x) - y(x)\|_2 \le L\sqrt{\rho_k(x)}. \tag{1.3}$$

*In particular, if identical signatures have a common deterministic label and $\rho_1(x) = 0$, the prediction is exact. For $T$ queries, $N$ training rows, feature dimension $d$, and target dimension $p$, the implemented full sort requires*

$$O(T[Nd + N \log N + Np]) \tag{1.4}$$

*arithmetic and $O(N(d + p) + TN)$ storage. A selection routine can replace the sort term by linear expected work but is not used in the reported code.*

*Proof.* A row permutation preserves the multiset of distances, the order statistic $\rho_k(x)$, and the set of target values satisfying the threshold; their mean is therefore unchanged. Under Eq. (1.2),

$$\begin{aligned} \|\widehat{y}(x) - y(x)\|_2 &\le \frac{1}{|\mathcal{N}_k(x)|} \sum_{i \in \mathcal{N}_k(x)} \|y(x_i) - y(x)\|_2 \\ &\le \frac{L}{|\mathcal{N}_k(x)|} \sum_{i \in \mathcal{N}_k(x)} \|x_i - x\|_2 \le L\sqrt{\rho_k(x)}. \end{aligned} \tag{1.5}$$

The exact-signature statement follows by setting the radius to zero. Computing all pairwise query–training distances costs $O(TNd)$, sorting costs $O(TN \log N)$, and averaging selected $p$-vectors costs at most $O(TNp)$. The stored data and distance array give the stated memory bound. □

**Table 1:** Mean held-out performance across four folds. The dash marks a method not evaluated under the stricter protocol. Angle MSE and relative QAOA regret are distinct estimands.

| Protocol | Method | Angle MSE | Relative regret |
|---|---|---|---|
| Row held out | constant | 0.09109 | 0.02042 |
| Row held out | ordinary nearest neighbor | 0.00137 | 0.00031 |
| Row held out | tie-complete neighbor | 0.00072 | 0.00031 |
| Row held out | kernel ridge | 0.00541 | 0.00800 |
| Row held out | rank-stability blend | 0.01438 | 0.01118 |
| Signature held out | constant | 0.07988 | 0.01643 |
| Signature held out | ordinary nearest neighbor | 0.00780 | 0.00267 |
| Signature held out | tie-complete neighbor | 0.00780 | 0.00267 |

This guarantee is local and conditional. The experiment therefore reports a second protocol in which every exact feature signature is assigned wholly to one fold. It measures performance when the zero-radius corollary cannot apply.

**2. Energy certificate.** For an unweighted graph $G = (V, E)$ with $n$ vertices and $m$ edges, let

$$C_G = \sum_{(u,v)\in E} \frac{I - Z_u Z_v}{2}, \qquad B = \sum_{v\in V} X_v. \tag{2.1}$$

At depth one [1],

$$F_G(\gamma, \beta) = \langle + | e^{i\gamma C_G} e^{i\beta B} C_G e^{-i\beta B} e^{-i\gamma C_G} | + \rangle. \tag{2.2}$$

**Theorem 2** (depth-one angle-to-energy bound)**.** *For $\theta = (\gamma, \beta)$ and $\theta' = (\gamma', \beta')$,*

$$|F_G(\theta) - F_G(\theta')| \le 2m^2|\gamma - \gamma'| + 2nm|\beta - \beta'|. \tag{2.3}$$

*If $F_G(\theta') > 0$, division by $F_G(\theta')$ gives a relative-regret certificate.*

*Proof.* For a unitary parameter generated by a Hermitian matrix $H$,

$$\left|\frac{\partial F_G}{\partial t}\right| \le \|[H, C_G]\|_2 \le 2\,\|H\|_2\,\|C_G\|_2\,. \tag{2.4}$$

Because $C_G$ is a sum of $m$ projectors, $\|C_G\|_2 \le m$; because $B$ is a sum of $n$ commuting Pauli operators, $\|B\|_2 \le n$. Integrating the two derivative bounds along the segment from $\theta$ to $\theta'$ proves Eq. (2.3). □

**3. Authenticated experiment.** G1 is an unweighted graph with 800 vertices and 19176 edges from the Stanford Gset collection [2]. Its bytes are accepted only after the registered SHA-256 digest matches. Forty-eight start vertices, fixed by $37s$ mod 800, define deterministic breadth-first ten-vertex induced neighborhoods. The maximum pairwise vertex-set Jaccard overlap is 0.25.

For each neighborhood, an exact 1024-amplitude statevector evaluator searches a fixed $9 \times 9$ grid:

$$\begin{aligned} \gamma &\in \mathrm{linspace}(0.08, \pi - 0.08, 9), \\ \beta &\in \mathrm{linspace}(0.04, \pi/2 - 0.04, 9). \end{aligned} \tag{3.1}$$

These labels are noiseless coarse-grid references rather than hardware measurements or continuous global optima. Eight graph descriptors yield 34 exact signatures; no signature contains conflicting reference labels.

Two four-fold evaluations are reported. The row protocol preserves the legacy index-modulo-four split and therefore permits repeated signatures across train and test sets. The signature protocol assigns every one of the 34 signatures to exactly one test fold; total train–test signature overlap is zero. This produces zero train–test signature overlap. Hyperparameters are selected inside each training partition. Table 1 reports both protocols under identical error and regret definitions.

Under row holdout, tie completion halves the neighbor angle MSE by averaging all zero-distance duplicates in one fold. It leaves mean regret unchanged. Under signature holdout, the two neighbor

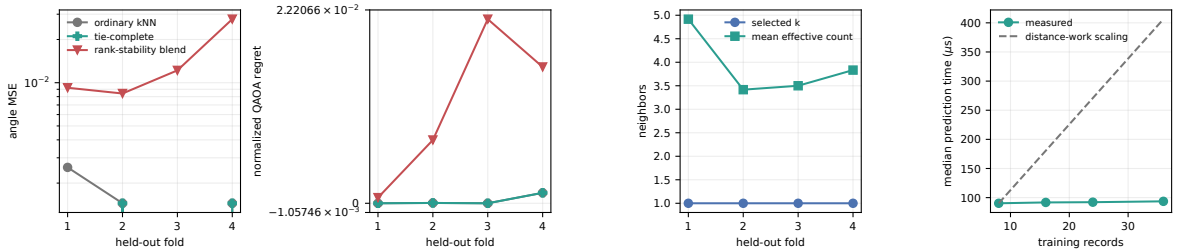



**Fig. 1:** Row-held-out audit. The first two panels separate angle error from energy regret. The third reports selected $k$ and the larger effective neighbor count caused by complete distance ties. The final panel times twelve real queries against prefixes of the actual training partition; fixed overhead dominates this small range despite the increasing distance work.

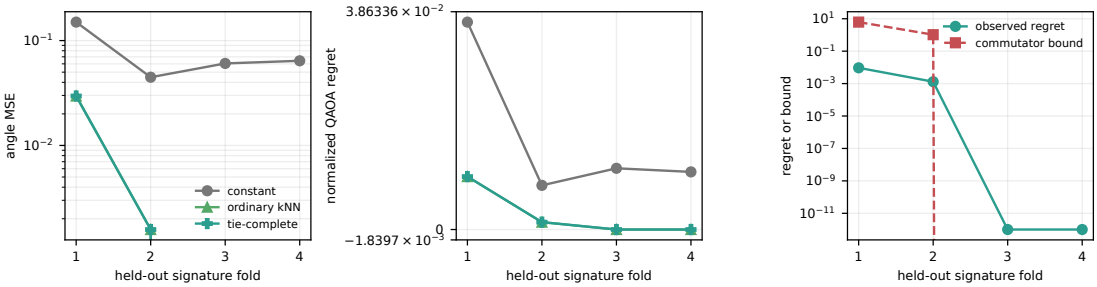



**Fig. 2:** Collision-free signature-grouped evaluation. Left and center: foldwise angle error and energy regret. Right: observed tie-complete regret and the commutator certificate from Theorem 2. The bound has no violations but is conservative.

methods agree to displayed precision. Their improvement over the constant predictor remains substantial, but the extrapolative MSE is more than ten times the row-held-out value. This gap is the central evidence boundary. Figure 1 diagnoses the favorable row-split result, whereas Fig. 2 exposes the collision-free generalization boundary and the theorem-based energy certificate.

**4. Interpretation and reproducibility.** The tie-complete rule solves a well-defined reproducibility defect: serialized row order cannot decide among equal-distance graph signatures. Its advantage on the row split is real but arises from exact-signature reuse. The signature-held-out experiment supplies no evidence that tie completion itself improves extrapolation; ordinary and tie-complete neighbors coincide there. Nor does a coarse-grid parameter label establish a continuous optimizer.

The package records G1 and G81 source addresses and byte hashes, all neighborhoods, grid settings, folds, signature groups, selected neighbor counts, effective tie counts, per-fold errors, energy certificates, timing samples, and locked outputs. The authenticated replay reconstructs all 48 landscapes, regenerates the tables and figures, and compares every non-timing value with the committed result. Tests additionally permute tied training rows and verify prediction invariance.

**References.**

[1] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm, 2014.

[2] Yinyu Ye. Gset: Test sets for the maximum cut problem. Stanford University, 2003. URL https://web.stanford.edu/~yyye/yyye/Gset/. Accessed July 31, 2026.